

1. Study Protocol: Update of Jones and DuVal (2019), and Davies et al. (2020)

1.1. Version history

Version number	Changes	Date	Responsible
1.11	Finalizing protocol for preregistration	01 Mar, 2026	Eduardo S. A. Santos

1.2. Title

Mate-choice copying in non-human animals: Protocol for the update of two meta-analyses.

1.3. Authors

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1.5. CRediT authorship contribution statement

- ESAS: Conceptualization, Methodology, Investigation, Data curation, Formal analysis, Visualization, Writing - original draft, Writing - review and editing, Project administration.

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- SN: Conceptualization, Methodology, Investigation, Writing - review and editing, Funding acquisition.
- ML: Conceptualization, Methodology, Investigation, Writing - review and editing.

1.6. Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Grammarly and Google Gemini to check grammar, spelling, and reference formatting. After using these tools, the authors carefully reviewed and edited the content as needed. The authors take full responsibility for the content of this report.

1.7. Data availability statement

All relevant data will be made available in a publicly accessible Borealis repository (<https://borealisdata.ca/>).

1.8. Competing interests statement

The authors have declared that no competing interests exist.

1.9. Funding

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2. Background

Social learning in mate choice is taxonomically widespread and may be a critical mechanism of social evolution, influencing the strength of sexual selection within populations. Mate-choice copying occurs when an individual's likelihood of mating with a specific individual of the opposite sex is influenced by the perceived choices of other members of the same sex.

Jones and DuVal (2019) published a comprehensive meta-analysis estimating the effect of female mate-choice copying on reproductive decisions. Their study evaluated 103 experiments from 40 original reports covering taxa such as Arachnida, Insecta, Malacostraca, Aves, and Actinopterygii. The authors found that females observing others were, on average, 2.71 times more likely to mate with the chosen male or a phenotypically similar individual. They also identified several key moderators. For instance, observer experience influenced copying decisions, with inexperienced (virgin) females more likely to make a copy choice. Mate-choice copying was more likely to occur when social information counteracted a female's personal or genetic preference (*i.e.*, favoring “unattractive” males). The environment in which females were also influenced mate-choice copying, with effect sizes being stronger in free-living subjects than in captive individuals.

Another meta-analysis conducted nearly simultaneously by Davies et al. (2020) also investigated mate-choice copying. However, this time included studies in which the focal/observer individuals could be male or female, thereby extending Jones and DuVal's (2019) study, which focused solely on female mate choice. Davies et al. (2020) reported strong evidence that mate-choice copying occurs in both males and females, and with a similar magnitude between the sexes. Jones and DuVal (2019) analysed 103 effect sizes from 40 studies, whereas Davies et al. (2020) analysed 158 effect sizes from 58 studies, a 53% increase in the number of effect sizes analysed. There is no information about the overlap in the evidence used between the two meta-analyses, but Davies et al. (2020) note in their supplemental material that their methods included more literature sources and screened 2.3 times more articles.

Foundational studies, such as the two almost-concurrent meta-analyses introduced above, should be reproduced, replicated, and updated. Reproducing a meta-analysis means using the same dataset as the original authors and the same code (or the same analytical steps) to verify accuracy and transparency. Replicating a meta-analysis involves testing the same research question using new data (for the same period or updated data) and/or a new search process, and employing new analytical procedures. Updating, as new evidence emerges, tests whether conclusions remain the same over time and draws conclusions from a larger, likely more diverse, evidence base. It also allows the application of newer, more robust analytical methods.

Testing reproducibility, replicating, and updating studies are fundamental parts of science (Nakagawa and Parker 2015). Scientists and the general public view meta-analyses as the gold standard for the quality of evidence on a specific question or research theme. As scientific outputs, published meta-analyses should be subject to replication and updating across all fields of knowledge. In fields such as the medical sciences, the discussion of whether evidence synthesis studies should be replicated has been addressed for over ten years. In medicine, the discussion now lies on the frequency and triggers that should prompt the replication and updating of meta-analytic studies (Group 2016). In ecology and evolution, only a few meta-analyses have been subjected to replication or update efforts (see, for example, Pollo et al., 2025). Given that several years have passed since the original literature searches in April (Davies et al.) and May (Jones and DuVal) 2019, an update is necessary to incorporate new empirical evidence and apply contemporary analytical methods.

2.1. Aims and questions

Based on (i) Jones and DuVal's (2019) and Davies et al.'s (2020) syntheses highlighting the importance of mate-choice copying for sexual selection theory, (ii) the amount of time since the publication of their meta-analysis, and (iii) the additional empirical studies that have been published since, our study aim to update Jones and DuVal's (2019) and Davies et al.'s (2020) meta-analyses on mate-choice copying by females.

Thus, our specific aim and research questions (Q) are:

- **Aim 1:** We will update Jones and Duval's (2019) and Davies et al.'s (2020) meta-analyses by incorporating new methods (search and analytical) and new data (from other bibliographic databases and multilingual studies).
 - Q1: Does the main finding of these two meta-analyses remain valid, or how much would they change if we apply the search strategy described in their reports to collect and update the datasets (information published since 2019), and use the same methods to analyze the data?
 - Q2: Does the main finding of these two meta-analyses remain valid, or how much would they change if we apply the search strategy described in their reports to collect and update the datasets (information published since 2019), and apply updated analytical methods?

Below, we present a diagram illustrating the combinations of search strategies and analytical methods that address our aim and questions (Table 1).

Table 1. Structured comparison of research approaches for addressing Q1 and Q2.

Elements	Q1:	Q2:
Search strategy	Same as in earlier meta-analyses (MA), applied only to search for studies published since 2019	More comprehensive search strategy (more databases and languages) with no time restriction
Included studies	Same as in earlier MA + new studies published since 2019	More included studies from all periods
Included effect sizes	Same as in earlier MA (no new extractions) + new effect sizes published since 2019 (new extractions)	More included studies from all periods - all extracted <i>de novo</i>
Analytical models	Same as in earlier MA	New analytical approaches
Overall:	Same methods, updating data by adding new studies to the existing datasets	Updated all methods and the whole dataset re-extracted and updated

3. Methods

3.1. Protocol and registration

This study protocol will be submitted for pre-registration on the Open Science Framework (OSF).

3.2. Ethics and dissemination

As this replication and update of a meta-analysis constitutes a summary and analysis of published literature, ethics approval is not required. The results of this study will be published in a peer-reviewed journal.

3.3. PECOS scope and eligibility criteria

The objectives of this study are to update Jones and DuVal's (2019) and Davies et al.'s (2020) meta-analyses on mate-choice copying behavior by non-human animals (see [Aims and questions](#), above).

Under the PECOS (Population, Exposure, Comparator, Outcomes, and Study design) framework (Richardson et al. 1995; Foo et al. 2021), our main eligibility criteria can be summarized as:

Population: Non-human, sexually mature female or male individuals, from wild or captive populations of multicellular animals.

Exposure: Tests of the audience effect during mate-choice trials, that is, the observation of a social cue (e.g., a female near or copulating with a male).

Comparator: A group (sample) of females or males that make a mate choice without the social cue or with a neutral/conflicting cue. For instance, random choice: females or males tested under identical conditions, either without a model female or male present or with the model female or male hidden by an opaque partition.

Outcome: The primary outcome is the mating preference of the focal/observer individual, typically measured by time allocation or copulation. For example:

- Preference Zone Time: The number of seconds spent within a specific distance of a male after the demonstration.
- Social Learning Index: A calculated value quantifying the bias toward the phenotype chosen by the model.
- Copulation Choice: Which phenotype the focal/observer individual actually chooses to mate with when physical access is allowed.

Study design: Eligible studies for inclusion in our project must be empirical experimental studies that expose focal/observer individuals to opposite-sex mates (with and without an audience effect).

We followed the inclusion criteria of Jones and DuVal (2019) and Davies et al. (2020), as well as the study scope described above. Our study inclusion criteria are as follows:

1. The study must be reported in English, Japanese, Polish, Portuguese, Russian, Simplified/Traditional Chinese, and Spanish;
2. The publication must be an empirical experimental study;
3. Study subjects must be non-human multicellular animals;
4. The study must investigate mate-choice copying by allowing one individual (focal/observer) to choose between two individuals of the opposite sex, one of which must be associated with stimuli from a model (demonstrator) individual of the same sex as the focal/observer;

5. Copying behavior must be investigated using some sort of audience effect (e.g., presence of a model/demonstrator individual, chemical, visual, or auditory cues, copulations);
6. Study must report individual, discrete choices; and
7. Studies must report effect size, count data, or significance statistics for the mate-choice copying trials.

3.3.1. Exclusion criteria

We will exclude studies according to the following criteria:

1. The study investigated only the behavior of observer/focal individual that belongs to the selection line, or is a mutant, or was physiologically manipulated, with no wild type results reported; and/or
2. The study employed complex manipulations of moderator variables, making it difficult to convert the reported results into effect sizes. An exception is when complex experimental studies provide raw data that allow us to calculate relevant effect sizes.

3.4. Information Sources

We will conduct a systematic search of academic literature to identify relevant studies for our update. Specifically, we will search the same databases used by the original meta-analyses: Web of Science Core Collection (hereafter referred to as Web of Science) and Scopus, with no restrictions on publication type or date, using the full search strings provided in the following section ([Search strategy](#)). In addition to the search databases used in the original meta-analyses, we will also search the Bielefeld Academic Search Engine (BASE), Scientific Electronic Library Online (SciELO), and Google Scholar. Following the search procedures of the two original meta-analyses, we will not impose any language restrictions on studies included in the literature databases, even if they are written in languages other than English. Specifically, on SciELO and Google Scholar, we will search for non-English-language literature (Japanese, Polish, Portuguese, Russian, Simplified/Traditional Chinese, and Spanish).

3.5. Search strategy

We conducted a pilot of our search string strategy to assess its sensitivity, feasibility, and transparency using the benchmarking/relative recall approach (Lagisz et al., 2025). We used a sample of 13 studies from Jones and DuVal's (2019) original selection as a benchmarking set (Applebaum and Cruz 2000; Danchin et al. 2018; Dugatkin and Godin 1993; Fowler-Finn et al. 2015; Freed-Brown and White 2009; Gierszewski et al. 2018; Howard et al. 1998; Matsumoto and Takegaki 2013; Milner et al. 2011; Monier

et al. 2019, 2018; Nöbel et al. 2018; Stiver and Alonzo 2011). This benchmark sample of references was created before any database searches were performed. We tried to encompass studies from all taxonomic groups and study types used in Jones and DuVal (2019). Below, we report the results of our pilot searches conducted in three databases: Web of Science, Scopus, and PubMed.

3.5.1. Web of Science

- Access date: 04 February 2026;
- Query: TS = (("mate choice" AND "copy*") OR ("mate-choice" AND "copy*") OR "mate-copy*" OR "mate copy*" OR "female* copy*");
- Total records retrieved: 474;
- Publication dates: 1900-2026; and
- Retrieval rate of 100%, with 13/13 benchmark references retrieved.

3.5.2. Scopus

- Access date: 04 February 2026;
- Query: TITLE-ABS-KEY (("mate choice" AND "copy*") OR ("mate-choice" AND "copy*") OR "mate-copy*" OR "mate copy*" OR "female* copy*");
- Total records retrieved: 1,228;
- Publication dates: 1900 to 2026; and
- Retrieval rate of 100%, with 13/13 benchmark references retrieved.

3.5.3. PubMed

- Access date: 05 February 2026;
- Query: ((("mate choice"[Title/Abstract] AND "copy*" [Title/Abstract]) OR ("mate-choice"[Title/Abstract] AND "copy*" [Title/Abstract]) OR "mate-copy*" [Title/Abstract] OR "mate copy*" [Title/Abstract] OR "female* copy*" [Title/Abstract])) AND ((("mate choice" AND "copy*") OR ("mate-choice" AND "copy*") OR "mate-copy*" OR "mate copy*" OR "female* copy*"));
- Total records retrieved: 174;
- Publication dates: 1900 to 2026; and
- Retrieval rate of 77%, with 10/13 benchmark references retrieved.

Our three pilot searches achieved very high retrieval rates, demonstrating that the search string we created is adequate to capture a representative sample of studies for our meta-analysis replication and update.

3.6. Study selection

To address Questions 1 and 2, we will select studies using the criteria described above (section [PECOS scope and eligibility criteria](#)). Once we retrieve lists of bibliographic records from Web of Science and Scopus using our search strategy, we will deduplicate records, filter those published since the beginning of 2019, and randomly assign them to our team for screening. For studies included in the second-stage screening (see below), we will seek full-text versions of the retrieved documents and extract data to calculate effect sizes.

3.7. Screening

To address Questions 1 and 2, we will conduct a two-stage screening procedure. First, we will screen the titles, abstracts, and keywords of the bibliographic references retrieved from our searches (after removing duplicate records). Then, for the subset of records that meet our first-stage inclusion criteria, we will conduct a full-text screening. Both screening stages will be performed using Rayyan (Ouzzani et al. 2016) as the platform. We will assess each record with at least two independent reviewers, using our decision trees (Figures 1 and 2 below) to ensure consistency in screening. For all records, a team member will be paired with either ESAS or ML as the second screener. Disagreements that arise during the screening stage will be resolved through discussion between the reviewers; if necessary, and if the disagreements persist, a third team member will act as a moderator.

We conducted a pilot screening procedure in which two of us, ESAS and ML, used the decision tree (Figure 1) to screen titles, abstracts, and keywords from a random sample of 100 records from our pilot Scopus search. We conducted the pilot screening in Rayyan and implemented the blind option, so we were unaware of each other's decisions. Our pilot resulted in an agreement rate of 94% in the first stage on whether to include or exclude studies in the second screening phase (full-text screening); only 6 studies required discussion to reach agreement. The screening pilot procedure took 40 minutes to screen 100 bibliographic records during the first stage. Our first-stage screening identified 19 papers that advanced to the next stage (full-text screening), yielding an inclusion rate of 19% among the studies identified in our search. For the second stage of our screening process (full-text screening), we both used the decision tree (Figure 2) to evaluate the 19 full-text articles in our sample; this took 30 minutes and resulted in a 100% inclusion rate. We only had to discuss the inclusion of 2% of the full texts, primarily because of the study's complex design. This led us to classify study designs by complexity as easy or hard, which will be one of the variables coded during data extraction.

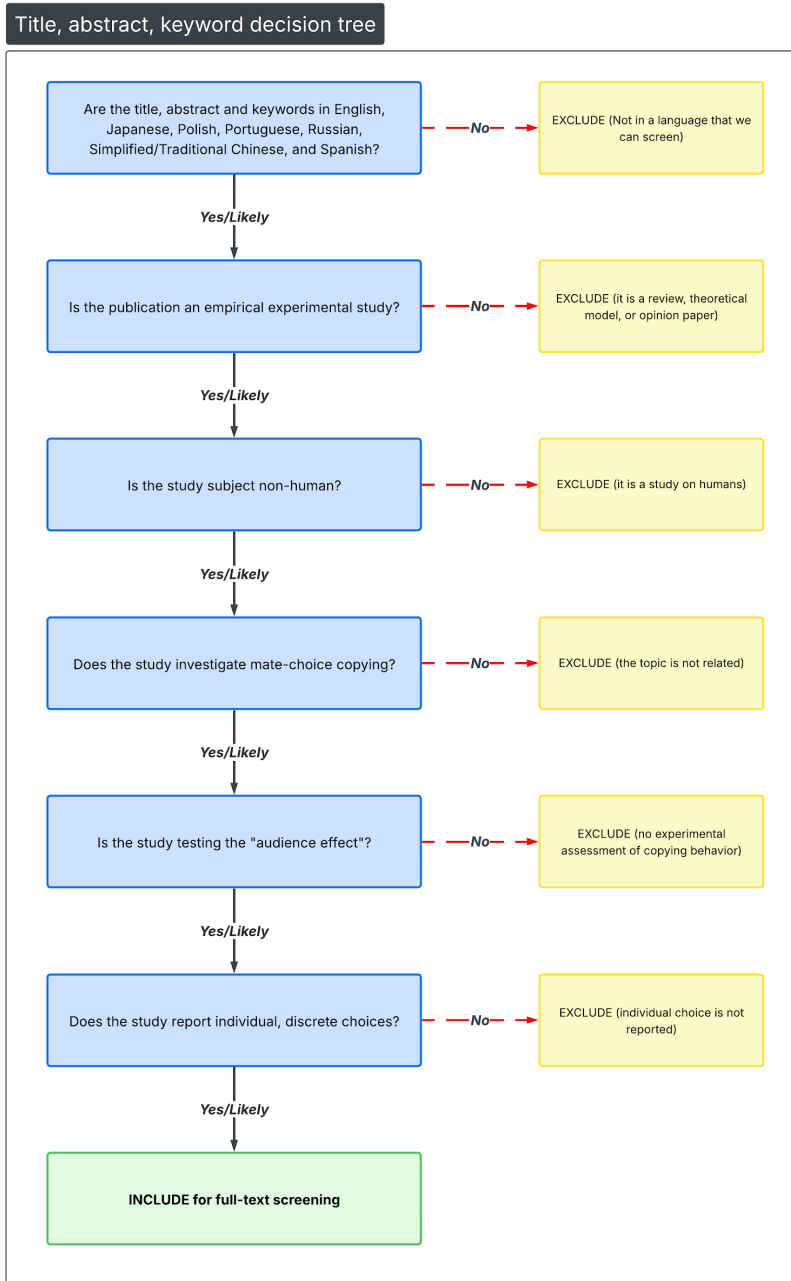


Figure 1. Screening decision tree for the title, abstract, and keywords (first-stage screening).

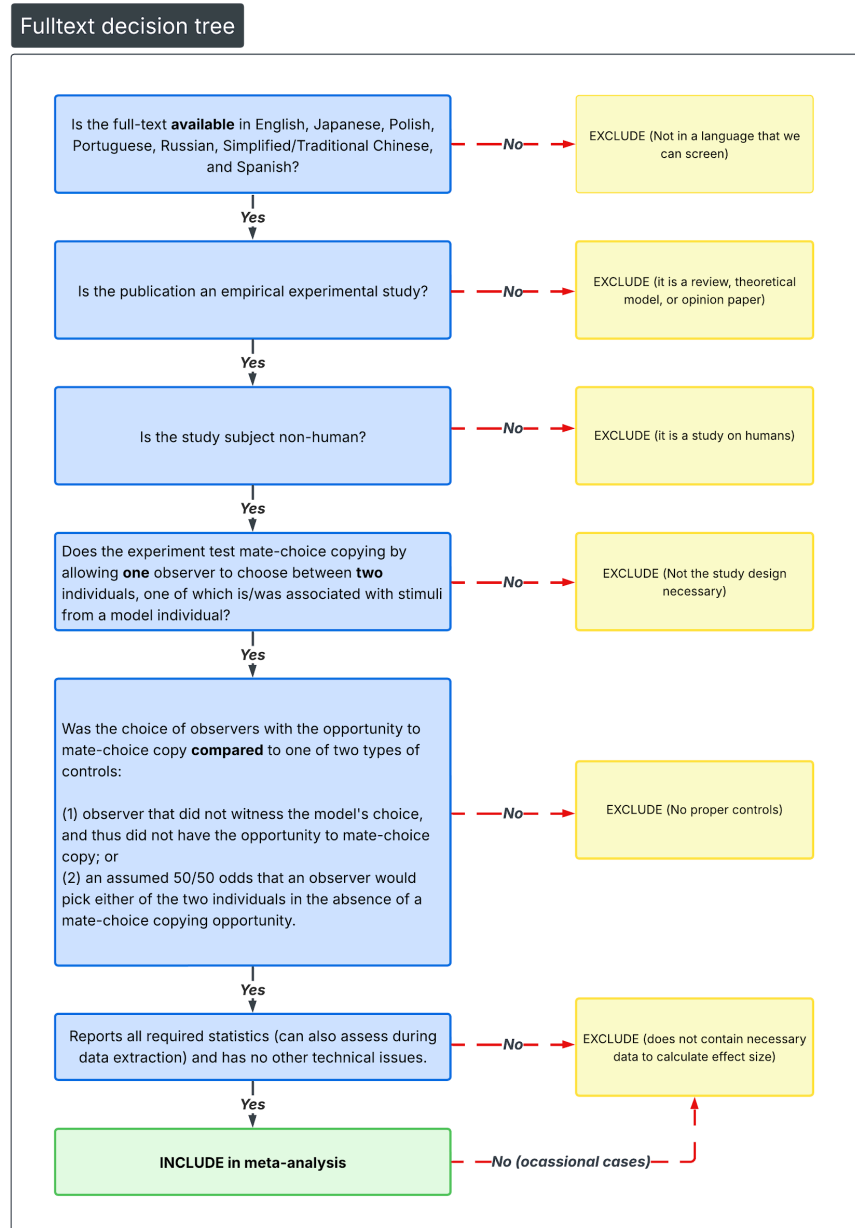


Figure 2. Screening decision tree for full texts (second-stage screening).

3.8. Data extraction

For Questions 1 and 2, we will read the full texts of the studies included in our screening and extract information from them when effect-size calculations are possible.

We will extract the data required to calculate effect sizes from text, tables, supplementary materials, and figures using the *metaDigitise* package (Pick et al., 2019). The variables to be extracted from each study are listed in Table 2. If a study reports multiple mate-choice copying trials, we will

extract information from all trials that meet our inclusion criteria. At least 10 percent of extracted records will be independently and blindly re-extracted by a second reviewer to assess agreement; discrepancies will be resolved through discussion.

Table 2. Data to be extracted from the empirical papers on mate-choice copying. Variables that cannot be extracted (due to unclear or missing information) will be labelled “NA”.

Variable	Description	Data type; options; examples of answers (separated by “/”)	Was this variable also used by Jones and DuVal (2019)?	Was this variable also used by Davies et al. (2020)?
identifierExtractor	Initials of the data extractor.	Restricted values; ; ES/SN/LL	No	No
identifierStudyId	The combination of the first author, the published year, and the published journal of the eligible study.	Free text; ; Santos_2006_The Journal	Yes	Yes
identifierStudyYear	Year of publication of the eligible study. Use the online publication year. If the online and print publication years differ, we will use the online publication year.	Numeric (four digits); ; 2006	Yes	Yes
identifierStudyDoi	Digital object identifier (DOI) of	Free text; ; 10.1098/rspb.2022	No	No

	the eligible study.	.2492		
taxonomySpecies	Species (binomial nomenclature)	Free text; ; Pan paniscus	Yes	Yes
experimentalSpeciesOrigin	Whether the study was conducted in the wild or in captivity. Studies in captivity maintain animals within facilities, whereas those in the wild use free-ranging animals that are not attended to.	Restricted values; ; Captivity/Wild	Yes	Yes
identifierExperimentId	Unique identifier for specific experiments within a paper. Experiment IDs within each study ID start with 001 and are sequential.	Restricted values; ; Exp001/Exp002/Exp003	Yes	Yes
experimentalDesignType	Experimental design type can be no pre-test, or before-and-after	Restricted values; ; No pre-test/Before-and-after	No	Yes
experimentalModelPairing	How model individuals were assigned during	Restricted; ; Manipulated/Natural	No	Yes

	the test. Manipulated pairing means that experimenters placed the demonstrator. Natural pairing means demonstrators were allowed to choose their partners.			
experimentalType eCopyingMechanism	Whether the observer was tested for Generalized Copying or Individual Copying. Generalized trials present observers with different target individuals that maintain a certain trait. Individual trials present observers with the same target individuals, just changing the audience effect.	Restricted; Generalized/Individual	No	Yes
experimentalType	Description of the	Free text; ; Actual	No	Yes

ChoiceTest	choice variable.	mating/Association time/Touch		
experimentalType TestDescription	Describe the type of mate-choice trial used by the author	Free text; ; “To test whether the presence of a female near a male influences the mate choice decision of another female, we gave 53 focal females a choice between two identical robotic crabs, one of which had a female tethered 5 cm away (to simulate a female visiting a male).”	No	Yes
experimentalType Sex	What was the sex of the observer/focal individual?	Restricted values; ; Female/Male	No	Yes
experimentalType Virgin	Is the focal/observer individual a virgin?	Restricted values; ; Yes/No	Yes	No
experimentalMod elAge	Age of the model/demonstrator or individual	Restricted values; ; Older/Same-age/Younger	Yes	No

	relative to the observer/focal.	ounger		
effectSizeSourceType	Source from where the effect size was extracted	Restricted values; ; Figure content/Figure legend/In-text/Table/Supplementary material	No	No
effectSizeSourceDetails	Details of the source from which the effect size was extracted	Free text; ; Figure 3a/Third paragraph in the 2nd page/Appendix 2	No	No
effectSizeEstimateType	Brief description of the estimate extracted. Prioritize data if presented as a sample size for each group of observer/focal individuals.	Free text; ; Count data/Correlation coefficient/t-value /p-value/Other	No	Yes
effectSizeEstimateValue	Value of the estimate extracted	Numeric; ; 0.5/200/1000	No	Yes
effectSizeEstimateExtra	Extra information about the estimate extracted, such as degrees of freedom or p-values	Free text; ; df = 40, p-value = 0.049	No	Yes

effectSizeEstimate Notes	Full description of the estimate extracted	Free text; ; Correlation coefficient is actually from a partial regression analysis	No	No
effectSizeSample SizeA	The number of observers choosing the individual associated with social information.	Numeric; ; 0.5/200/1000	No	No
effectSizeSample SizeB	The number of observers choosing the individual NOT associated with social information.	Numeric; ; 0.5/200/1000	No	No
effectSizeSample SizeC	The number of control individuals (i.e., individuals that did not witness the model's choice) that made the same choice as the model.	Numeric; ; 0.5/200/1000	No	No
effectSizeSample SizeD	The number of controls that did	Numeric; ; 0.5/200/1000	No	No

	not choose the same as the model.			
effectSizesampleSize	Sample size as the number of individuals. This is the total sample size grouping all focal/observer individuals used in the trial.	Numeric; ; 15/42/73	No	Yes
descriptionStudyComplexity	Level of the complexity of the study design and analytical methods - to be used as a proxy of effort/expertise needed to extract the data.	Restricted values; ; Easy/Moderate/Hard	No	No
descriptionGeneralNote	Relevant notes on the extraction.	Free text; ; I don't know if this metric can be used to calculate an effect size.	No	No

3.9. Data synthesis

Most studies in our dataset likely report effect sizes as either the number of observers who chose a specific individual (2 x 2 contingency tables) or as inferential statistics. Following Jones and DuVal

(2019), we will use odds ratios (OR) as the effect size to update their study. For our analyses, we will transform OR to $\log(\text{OR})$.

It is important to note that Davies et al. (2020) used Hedges' d as the effect size in their meta-analysis. Thus, to update Davies et al. (2020), we will calculate Hedges' d . Hedges' d can be converted into OR values, and vice-versa, so we can change effect sizes from one metric to the other when needed.

For Q1, we will follow the analytical methods described in Jones and DuVal (2019) and Davies et al. (2020) to apply meta-analytical methods. To address Q2, using updated analytical methods, we will employ multilevel meta-analytic models with empirical study ID, species ID, and phylogenetic relatedness as random effects to control for non-independence. We will quantify various heterogeneity metrics (e.g., I^2 , CV , M ; see Yang et al., 2025) and the proportion of the total heterogeneity explained by each random factor (e.g., partial I^2). However, we might remove random effects that account for little of the heterogeneity in effect sizes (i.e., when partial $I^2 < 0.1\%$). We may also conduct meta-regressions if the data distribution permits.

To compare estimates from the original meta-analysis with our new estimates, we will assess them qualitatively and quantitatively. We will first determine whether an estimate is positive, negative, or not different from zero (if its 95% CI overlaps zero), so that estimates with distinct classifications represent a qualitative difference. By contrast, a quantitative difference is defined as an absolute difference between two estimates that yields a z -score greater than 1.96 (i.e., statistically significant at two-tailed $\alpha = 0.05$). This approach follows the procedure used by Pollo et al. (2025) to evaluate the reliability of meta-analyses in ecology and evolutionary biology.

3.10. Publication bias

We will assess for publication bias in our meta-analyses using three approaches. First, we will visually inspect funnel asymmetry using a funnel plot, plotting the residuals from our global meta-analytic models against the precision of effect sizes (i.e., the inverse of the standard error). Second, we will use the inverse of the effective sample size as a moderator in multilevel meta-analytical models (Nakagawa et al. 2022). Third, we will verify time-lag bias by including publication year and using the original versus new timespan of evidence as a categorical variable (i.e., old vs. new data) as a moderator in a multilevel meta-analytic model.

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